Review Questions for Lecture 18

In questions 1-3, imagine that you are on a game show and you win a prize if you correctly guess a secret number x, chosen by the game show host. In each scenario below, the game show host will give you a clue to help you guess x. What strategy should you use to guess x?

1.The game show host flips a fair coin 100 times and counts the number of heads. She adds this to x and tells you the result.

a. If you can make three guesses, which three numbers would you pick?

b. Using your strategy, what is the chance you will win?

2. The game show host rolls a fair six-sided die, adds to x the number that comes up, and tells you the result.

a. If you can make three guesses, which three numbers would you pick?

b. Is your strategy optimal, or are there other strategies that are just as good?

c. Using your strategy, what is the chance you will win?

3. The game show host flips a coin 100 times and counts the number of heads. She takes the number of heads, multiplies it by two, adds x, and tells you the result.

a. If you can make three guesses, which three numbers would you pick?

b. Using your strategy, what is the chance you will win?

4. For each of the following, say whether it is random or fixed, and explain briefly.

a. the average of the box,

b. the confidence interval,

c. the standard error of the sample average,

d. the sample,

e. the sample average,

f. the expected value of the sample average,

g. the standard deviation of the box,

h. the standard deviation of the sample,

i. the bootstrap estimate of the standard deviation of the box, and

j. the bootstrap estimate of the standard error of the sample average.

5. Usually when we talk about 95% confidence intervals, we actually mean **approximate** 95% confidence intervals. In other words, the probability that a confidence interval will end up including the parameter isn't **exactly** 95%, just approximately. The reason for this is that we make two approximations when we construct the confidence interval. What are the two approximations?